

1. (Fall 2024, #8) Use results from the course to show that $f(x) = x^3 + 25x + 8$ has exactly one real root by showing the following steps. If a theorem is required, make sure to briefly, but explicitly, verify the hypotheses of the theorem.

(a) Show that $f(x) = x^3 + 25x + 8$ has at least one real root.

$$f(1) = 1 + 25 + 8 = 34 > 0$$

$$f(-1) = (-1)^3 + 25(-1) + 8 = -1 - 25 + 8 = -18 < 0$$

Since $f(x)$ is continuous, we can apply IVT.

By IVT $\Rightarrow$ there is some $x \in (-1, 1)$ such that $f(x) = 0$.

(b) Show that $f(x) = x^3 + 25x + 8$ has at most one real root.

Recall: $\left\{ \begin{array}{l} \text{If } f(x) \text{ is a differentiable function with } \underline{n} \text{ real roots, then } f'(x) \text{ has at least } \underline{n-1} \text{ real roots.} \\ \text{If } f(x) \text{ is a differentiable function and } f'(x) \text{ has } \underline{n} \text{ real roots, then } f(x) \text{ has at most } n+1 \text{ real roots.} \end{array} \right.$

Since these are the consequences of MVT, from now on I will refer to them as MVT.

$$f'(x) = \underbrace{3x^2}_{\geq 0} + \underbrace{25}_{> 0} > 0 \Rightarrow f'(x) \text{ has } 0 \text{ real roots.}$$

$\xrightarrow{\text{MVT}} f(x) \text{ has at most } \underline{0+1} \text{ real roots.}$
 $\uparrow$
 $\underline{1}$

We can apply MVT since $f(x)$ is differentiable.

2. (Spring 2024, #5) Consider the function over the interval $(0, 2)$ defined by:

$$f(x) = \frac{1}{x} - \frac{1}{x-2}$$

(a) What are $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$?

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} - \frac{1}{x-2} = +\infty$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{1}{x} - \frac{1}{\underbrace{x-2}_{-\infty}} = +\infty$$

(b) Show that f is concave up over the interval $(0, 2)$.

Concave up: $\begin{cases} f'(x) \text{ is increasing on } (0, 2) \\ f''(x) \geq 0 \text{ on } (0, 2) \end{cases}$ Since $0 < x < 2 \Rightarrow \begin{cases} 0 < x \Rightarrow x^3 > 0 \\ x-2 < 0 \Rightarrow (x-2)^3 < 0 \end{cases}$

$$f'(x) = -\frac{1}{x^2} + \frac{1}{(x-2)^2} = -x^{-2} + (x-2)^{-2} \Rightarrow f''(x) = -(-2)x^{-3} - 2(x-2)^{-3} = \frac{2}{\underbrace{x^3}_{>0}} - \frac{2}{\underbrace{(x-2)^3}_{<0}} > 0$$

(c) Use an appropriate theorem to show that $f'(x) = 1$ has at least one solution over the interval $(0, 2)$.

(We can use IVT since $f'(x)$ is continuous.)

$$\begin{aligned} \Rightarrow f'\left(\frac{1}{2}\right) &= -\frac{1}{\left(\frac{1}{2}\right)^2} + \frac{1}{\left(-\frac{3}{2}\right)^2} = -4 + \frac{4}{9} < 1 \\ f'\left(\frac{3}{2}\right) &= -\frac{1}{\left(\frac{3}{2}\right)^2} + \frac{1}{\left(-\frac{1}{2}\right)^2} = -\frac{4}{9} + 4 > 1 \end{aligned}$$

IVT $\Rightarrow$ there is some $\frac{1}{2} < x < \frac{3}{2}$ such that $f'(x) = 1$.

other way: $\left(\text{check: } \lim_{x \rightarrow 0^+} f'(x) = -\infty \text{ and } \lim_{x \rightarrow 2^-} f'(x) = +\infty \text{ and then use IVT} \right)$

(d) Use part (b) to show that $f'(x) = 1$ has at most one solution in the interval $(0, 2)$.

Let $g(x) = f'(x) - 1$. Then $g'(x) = f''(x)$

If $g(x) = f'(x) - 1$ has 0 real roots, then $g(x) = f'(x) - 1$ has at most 1 real root.

In part (b) we showed $f''(x) > 0 \Rightarrow f''(x)$ has no real roots.

MVT $\Rightarrow f'(x) - 1$ has at most 1 real root.

$\uparrow$
we can apply MVT because $f'(x) - 1$ is a differentiable function on the interval $(0, 2)$.

3. (Spring 2023, #5) Consider the equation $e^x \cdot x^2 = 1$.

- (a) Show that the equation has at least one solution on the interval $[0, \infty)$ and justify your work by citing the appropriate theorem(s) from this course.

$$f(x) = e^x \cdot x^2 - 1$$

$$f(1) = e^1 \cdot (1)^2 - 1 = e - 1 > 0$$

$$f(0) = e^0 \cdot (0)^2 - 1 = -1 < 0$$

$\Rightarrow$
IVT

There is some x in $[0, 1)$

such that $f(x) = 0$

which means $e^x \cdot x^2 = 1$

(We can apply IVT
because $f(x)$ is continuous)

- (b) Show that the equation has at most one solution on the interval $[0, \infty)$ and justify your work by citing the appropriate theorem(s) from this course.

$$f(x) = e^x \cdot x^2 - 1$$

$$f'(x) = e^x \cdot x^2 + 2x \cdot e^x$$

We can apply MVT because $f(x)$ is differentiable.

$$f'(x) = 0 \quad \text{for } x = 0$$

$f'(x) > 0$ for $x > 0 \Rightarrow$ Since $f'(x)$ has no real roots on the interval $(0, \infty)$, $f(x)$ has at most 1 real root on the interval $[0, \infty)$.

Indeed if $f(x)$ had two real roots on $[0, \infty)$, then f' should have 1 real root greater than zero (between the two real roots of $f(x)$).

4. (Fall 2022, #4)

(a) Show that the equation $\cos(x) + 3x = 0$ has exactly one root.

Let $f(x) = \cos(x) + 3x$.

$$f(1) = \cos(1) + 3 > 0 \quad (\text{since } -1 < \cos(1) < 1)$$

$$f(-1) = \cos(-1) - 3 < 0$$

$\Rightarrow f(x)$ has at least one real root. (I)

IVT

$\uparrow$

(we can apply IVT because $f(x)$ is continuous.)

$$f'(x) = -\sin(x) + 3$$

Again since $-1 \leq -\sin(x) \leq 1 \Rightarrow 2 \leq -\sin(x) + 3 \leq 4$ which is always

Positive. So $f'(x)$ has no real roots $\Rightarrow f(x)$ has at most 1 real root (II)

MVT (because $f(x)$ is differentiable)

I $\Rightarrow$ $f(x)$ has exactly 1 real root.

(b) Find the absolute maximum and absolute minimum of $f(x) = \frac{x}{(x^2 - x + 1)}$ as x ranges over $[0, 3]$.

Hint: $f'(x) = \frac{(1-x)(1+x)}{(x^2 - x + 1)^2}$

Typo: $f(x) = \frac{x}{x^2 - x + 1}$

Critical points:

$$f'(x) = 0 \Rightarrow (1-x)(1+x) = 0 \Rightarrow x = 1 \text{ or } x = -1. \quad (\text{We only consider } x=1, \text{ since } -1 \notin [0, 3])$$

$$f'(x) \text{ DNE} \Rightarrow (x^2 - x + 1)^2 = 0 \Rightarrow x^2 - x + 1 = 0 \Rightarrow x^2 - x + \frac{1}{4} + \frac{3}{4} = 0 \Rightarrow (x - \frac{1}{2})^2 + \frac{3}{4} = 0$$

which has no solution.

$$\text{endpoints } \begin{cases} f(0) = \frac{0}{0-0+1} = 0 \\ f(3) = \frac{3}{3^2-3+1} = \frac{3}{7} \end{cases}$$

$$\text{critical points: } f(1) = \frac{1}{1^2-1+1} = \frac{1}{1} = 1$$

$$\Rightarrow \begin{cases} \text{absolute max: } x = 1 \\ \text{absolute min: } x = 0 \end{cases}$$

5. (Fall 2021, #5) Find the exact number of solutions to the equation $4 - x = e^x$. Justify your conclusion by citing appropriate theorem(s) from this course.

Let $f(x) = e^x + x - 4$

since $f(x)$ is continuous.

$$f(0) = e^0 + 0 - 4 = -3 < 0$$

$\xRightarrow{\text{IVT}}$ $f(x)$ has at least 1 root on the interval $(0, 2)$

$$f(2) = e^2 + 2 - 4 = e^2 - 2 > 0$$

Since $f(x)$ is differentiable.

$f'(x) = e^x + 1 > 0 \Rightarrow f'(x)$ has no real roots $\xRightarrow{\text{MVT}}$ $f(x)$ has at most 1 real root.

$\textcircled{\text{I}} \Rightarrow f(x)$ has exactly 1 real root

$\textcircled{\text{II}}$

$\Rightarrow e^x = 4 - x$ has exactly 1 real solution!